28/11/2017
Evening

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam

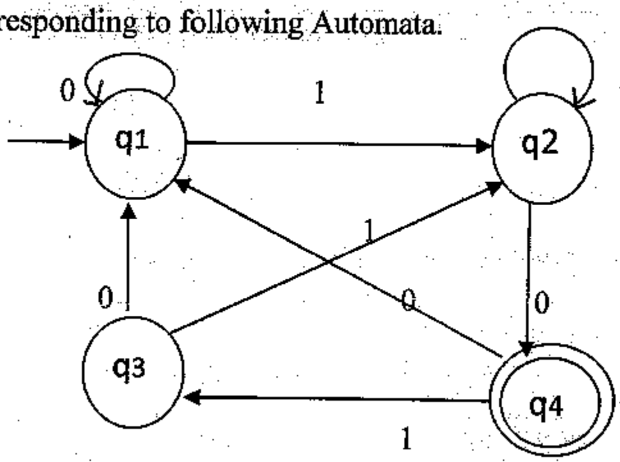
NOV-DEC 2017

Max. Marks: 100
Class: SE
Name of the Course: Theoretical Computer Science
Branch: Computer
Course Code: UCEC405

Duration: 3 hrs
Semester: MU-CBGS-IV

Instructions:
(1) All Questions are Compulsory
(2) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	Design a DFA to accept a] a set of strings which begin and end with different letters over Σ={x,y,z} b] a set of all strings with odd number of one's followed by even number of zeros. (Show Transition diagram, Table and tuple details) **OR**	10
Q 1 (a)	Design FSM to check whether given ternary number is divisible by 5. Simulate behavior of FSM for some valid string and invalid string.	10
Q 1 (b)	Construct an NFA with epsilon transitions (i) (00 + 1) * (10)* (ii) ((0 + 1)* 10 + (00)*(11)*)* Convert NFA for **(00 + 1) * (10)*** to its equivalent DFA	10
Q2 (a)	Differentiate between Moore and Mealy machine. Design mealy machine to add two binary numbers of equal length.	10
Q2 (b)	Design Moore machine and Mealy machine to get 2's compliment of a binary number	10
Q3 (a)	Check the following grammar for ambiguity. Remove ambiguity if any. E → E+E / E*E / E-E / E/E / (E) / a. Generate parse tree for (a+a)*(a-a) **OR**	10
Q 3 (a)	Prove that the following language is not a context free using Pumping Lemma, L={ $a^n b^n c^n$ / n>=0}	10

Q3 (b)	Write CFG for, A] $L=\{ a^n b^n 2^i \ / \ n, i>=1\}$ B] Strings containing alternating sequence of 0's and 1's over $\Sigma=\{0,1\}$	10
	OR	
Q 3 (b)	Write regular expression for the following language a) All strings containing not more than two a's over the $\Sigma=\{a,b\}$ b) $L=\{ w \ / \$ number of a's in w must be divisible by 3, $\Sigma=\{a,b\} \}$, $W \in \{a,b\}^*$ c) $L= \{ w \ / \ \|w\| \bmod 3 =0$, over the $\Sigma=\{0,1\}$, $\omega \in \{a,b\}^*$ d)all strings with no consecutive 1's over $\Sigma=\{0,1\}$	10
Q4 (a)	Design PushDown Automata for well balanced parenthesis. $\Sigma=\{ (,), \{, \} \}$. Show simulation for "({ }) ( ) "	10
	OR	
Q 4 (a)	Design PDA that accepts the following language. a] $L= \{ WcW^R \ / \ W \ \varepsilon \ \{a,b\}^*$ and c is any terminal) Show simulation for the string "abbcbba"	10
Q4 (b)	State Arden's Theorem and use it to construct regular expression coreresponding to following Automata. 	10
Q5 (a)	Design Turing Machine for, $L=\{ a^n b^{2n} \ / \ n>=1\}$. Show simulation for "aabbbb".	10
Q5 (b)	Explain Halting Problem and Post correspondence Problem	10

16.12.2017 (E)

K. J. Somaiya College of Engineering, Mumbai-77

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End Semester Exam

NOV- DEC 2017

Max. Marks: 100 | **Duration: 3 Hrs**

Class: S Y BTech | Semester: IV

Name of the Course: Applied Mathematics-IV | Branch: COMP/IT

Course Code: UCEC401 / UITC401

Instructions:

(1) All Questions are Compulsory

(2) Figures to right indicate full marks. Each sub-question has equal marks.

Question NO.		Questions	Marks
Q.1	A	Fit a Parabola to the following data and estimate $y(2.4)$ x: 1, 2, 3, 4 y: 1.7, 1.8, 2.3, 3.2	5
	B	**Attempt any Two of the following.**	16
	(i)	Obtain the rank correlation coefficient from the following data. X : 10, 12, 18, 18, 15, 40. Y : 12, 18, 25, 25, 50, 25.	
	(ii)	The regression lines of a sample are $x + 6y = 6$, and $3x + 2y = 10$. Find **(a)** Sample means $\bar{x}$ and $\bar{y}$, **(b)** coefficient of correlation between x and y. **(c)** Estimate y when $x = 12$.	
	(iii)	Obtain two lines of regression and coefficient of correlation from the following data x: 65, 66, 67, 67, 68, 69, 70, 72 y: 67, 68, 65, 66, 72, 72, 69, 71	

Table for Q.1 A:

x	1	2	3	4
y	1.7	1.8	2.3	3.2

Table for Q.1 B (iii):

x:	65	66	67	67	68	69	70	72
y:	67	68	65	66	72	72	69	71

Q.2	A	Let the mileage (In thousand of miles) of a particular tire be a random variable X having the probability density $f(x) = \begin{cases} \frac{1}{20} e^{-x/20}, & \text{if } x > 0 \\ 0, & \text{if } x \leq 0 \end{cases}$ Find the probability that one of these tires will last **(a)** at most 10,000 miles **(b)** anywhere from 16,000 to 24,000 miles	5
	B	**Attempt any Two of the following.**	16
	(i)	In a normal distribution exactly 7% of items are under 35 and 89% are under 63. What are the mean and standard deviation?	
	(ii)	An irregular six faced dice is thrown. The probability that in 10 throws it will give five even numbers is twice as likely that it will give four even numbers. How many times in 10,000 sets of 10 throws, would you expect to give no even number.	
	(iii)	After correcting 50 pages of the proof of a book, the reader finds that there are on the average 2 errors per 5 pages. How many pages would one expect to find 0, 1, 2 and 3 errors in 1000 pages of first print of the book?	
Q.3	A	An ambulance service claims that it takes on an average 8.9 min to reach the destination in emergency calls. To check this the Licensing Agency has then timed on 50 emergency calls, getting a mean of 9.3 min with a S.D. 1.6 min. Is the claim acceptable at 5% LOS?	5
	B	**Attempt any Two of the following.**	16
	(i)	Ten accountants were given intensive coaching and tests were conducted before and after coaching. The scores of tests are given below.	

Sr. No	1	2	3	4	5	6	7	8	9	10
Marks in test before coaching	50	42	51	42	60	41	70	55	62	38
Marks in test after coaching	62	40	61	52	68	51	64	63	72	50

Does the score show an improvement? Test at 5% level of significance

	(ii)	Based on the data below determine if there is relation between literacy and smoking:	
	(iii)	Six guinea pigs injected with 0.5 mg of a medication took on an average 15.4 secs. to fall asleep with an unbiased standard deviation 2.2 secs., while six other guinea pigs injected with 1.5 mg. of the medication took over an average 11.2 secs. to fall asleep with an unbiased standard deviation 2.6 cms. Use 5% level of significance to test the null hypothesis that the difference in dosage has no effect	
Q.4		**Attempt any TWO of the following.**	16
	(i)	A repair shop, attended by a single mechanic, has an average of four customers an hour who bring small appliances for repair The mechanics inspects them for defects and for this he takes six min. on an average. Arrivals are Poisson and service rate has an exponential distribution. You are required to: **(a)** Find the proportion of time during which there is no customer in the shop **(b)** Find the probability of finding at least one customer in the shop **(c)** Calculate the average number of customers in the system **(d)** Find the average time spent by a customer in the shop including service	
	(ii)	A Xerox machine owner earns by giving Xeroxing services.The time required to complete Xeroxing of one customer has an exponential distribution with the mean of 5 minutes.. The arrival of customers is a Poisson process with mean rate of 6 customers an hour. If the machine ownwr works 8 hours a day, find (i) the percentage idle time, (ii) the average time a customer has to remain in the shop, (iii) the average number of customers in the queue, (iv) the probability that there will be more than 4 customers in the shop.	
	(iii)	At a railway station, only one train is handled at a time the railway yard is sufficient only for three trains . Trains arrive at the station at an average rate of 6 per hour and the railway station can handle them on an average 12 per hour. Assuming Poisson arrivals and exponential service distribution, find **(a)** the steady state probabilities for the various number of trains in the system. **(b)** the average time spent by a train at the yard. **(c)** the average number of trains at the yard **(d)** the probability of having 2 trains at the yard	

Table for Q (ii):

	Smokers	Non-smokers
Literates	83	57
Illiterates	45	68

Q.5	A	Find the Maximum or minimum of the function $Z = x_1^2 + x_2^2 + x_3^2 - 6x_1 - 8x_2 - 10x_3$	5
	B	**Attempt any Two of the following.**	16
	(i)	Solve the following LPP by simplex method. Maximise $Z = 100x_1 + 50x_2 + 50x_3$ Subject to $4x_1 + 3x_2 + 2x_3 \leq 10,\ 3x_1 + 8x_2 + x_3 \leq 8,\ 4x_1 + 2x_2 + x_3 \leq 6$ $x_1, x_2, x_3 \geq 0$	
	(ii)	Using the method of Lagrange's multipliers solve the following problem: Optimize $z = x_1^2 + x_2^2 + x_3^2$ Subject to $x_1 + x_2 + 3x_3 = 2, 5x_1 + 2x_2 + x_3 = 5, x_1, x_2, x_3 \geq 0$	
	(iii)	Using the penalty (Big M) method solve the following LPP. *Minimise* $z = 2x_1 + 3x_2$ *subject to* $x_1 + x_2 \geq 5$ $x_1 + 2x_2 \geq 6$ $x_1, x_2 \geq 0$	

12.12.2017 (E)

K. J. Somaiya College of Engineering, Mumbai-77
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End Semester Exam
Nov - Dec 2017

Max. Marks:100
Class:SE
Name of the Course:Relational Database Management Systems(RDBMS)
Course Code: UCEC404

Duration: 3 Hours
Semester:IV
Branch: COMP

Instructions:
(1) **All Questions are Compulsory**
(2) **Draw neat diagrams .**
(3) **Assume suitable data if necessary**

Question No.		Max. Marks
Q 1 (a)	Explain difference between decomposition and normalization. Explain n 1NF, 2NF and 3NF giving examples.	10
Q 1 (b)	Consider a relation 'R' with attributes A,B,C,D,E. R={A,B,C,D,E}. The following dependencies hold; A→B BC→E ED→A. i) List all keys ii) is R in 1NF iii) Is R in BCNF	10
Q2 (a)	Give the advantages of DBMS over file system. Explain the role and responsibilities of database administrator.	10
Q2 (b)	Construct an ER diagram for a car insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents.	10
	OR	
	Explain the following with proper example i) Weak entity set ii) Total and partial participation iii) Mapping cardinality iv) Aggregation v) Specialization	10
Q3 (a)	Explain the following Relational Algebraic operations with examples :- (i) Generalized Projection (ii) Set Intersection (iii) Aggregate (iv) Natural Join (v) Assignment.	10
Q3 (b)	Explain the steps for converting ER Data model to Relational Data Model with proper examples.	10
	OR	
	What do you mean by Authorization and Authentications in DBMS? Explain how it is implemented in SQL with suitable example.	10

Q4 (a)	Consider the following relations scheme Emp(eid,ename,age,salary) Works(eid,did,pct_time) Dept(did,budget,managerial) (Employees can work in more than one department and pct_time gives % of time in that dept.) Write the following queries in SQL i) Print name and salary of employees whose salary exceeds the budget of all departments where he / she works in. ii) Find names of managers who manage most number of departments. iii) Find top 10 earning employees. iv) Delete all employees, who do not put in 25% work in atleast 1 department in which they work in.	10
Q4 (b)	What is view in SQL, how it is defined? Discuss the problem that may arise when we attempt to update a view. How views are implemented.	10
	OR	
	Explain the following with proper example i) Assertion ii) Trigger	10
Q5 (a)	Draw and explain state transition diagram of transaction. Explain ACID property of transaction.	10
Q5 (b)	What is deadlock? What are the conditions necessary for deadlocks? Discuss the various strategies used for handling deadlock.	10
	OR	
	What is recoverable schedule? Why is recoverability of schedules desirable? Are there any circumstances under which it would be desirable to allow non-recoverable schedules? Explain.	10